



lca_supplymat

A new brightway LCA method
to quantify manufactured
materials from inventory data

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SUMMARY

- The need
- The solution
- Principle
- Setup
- Live demo on **Activity Browser**

THE NEED

Context :

- **RTE** : French Transmission System Operator
- **Project** : Prospective LCA of the french electricity mix

Need : Evaluate gross usage of materials (metals, concrete, ...)

Constraints :

- Not only metals / biosphere extraction :
Can't use Relics
- Compatibility with legacy *Brightway* / *Ica_algebraic* / *Activity Browser* :
Can't use EDGES

THE SOLUTION

Create a **pseudo** LCIA method for Brightway, for each material.

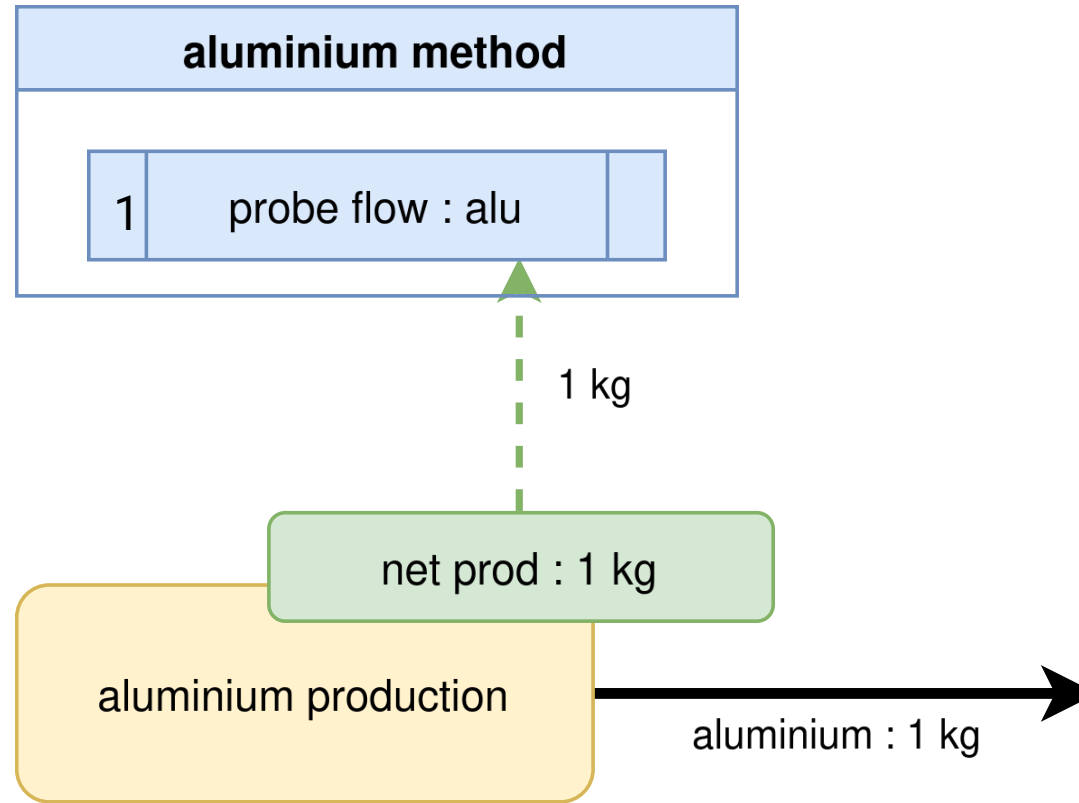
Matching on *reference products*, not activity name.

Add a "biosphere" database with **pseudo probe elementary** flows.

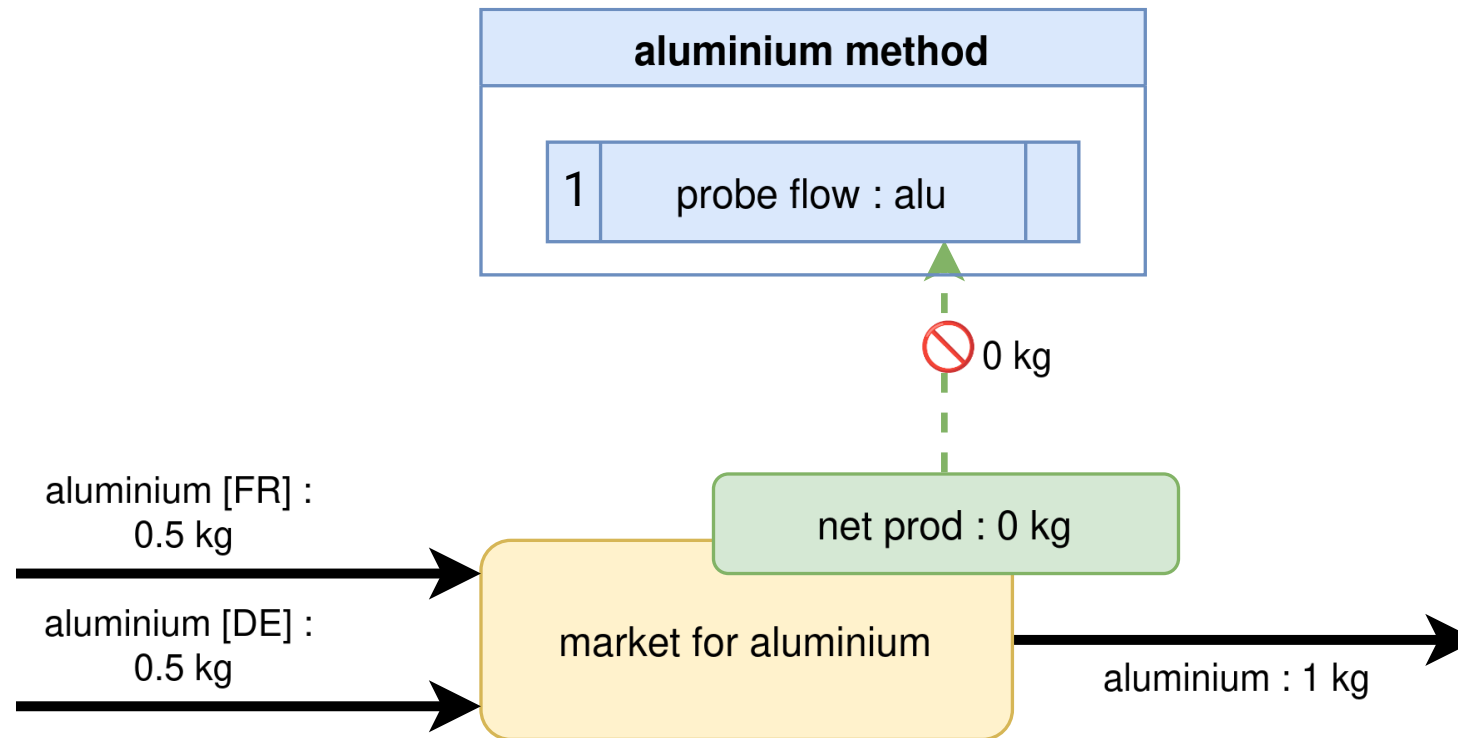
Add **custom exchanges** targeting **pseudo biosphere flows**, on each matching activity.

Terminology : *pseudo* LCIA and biosphere flows

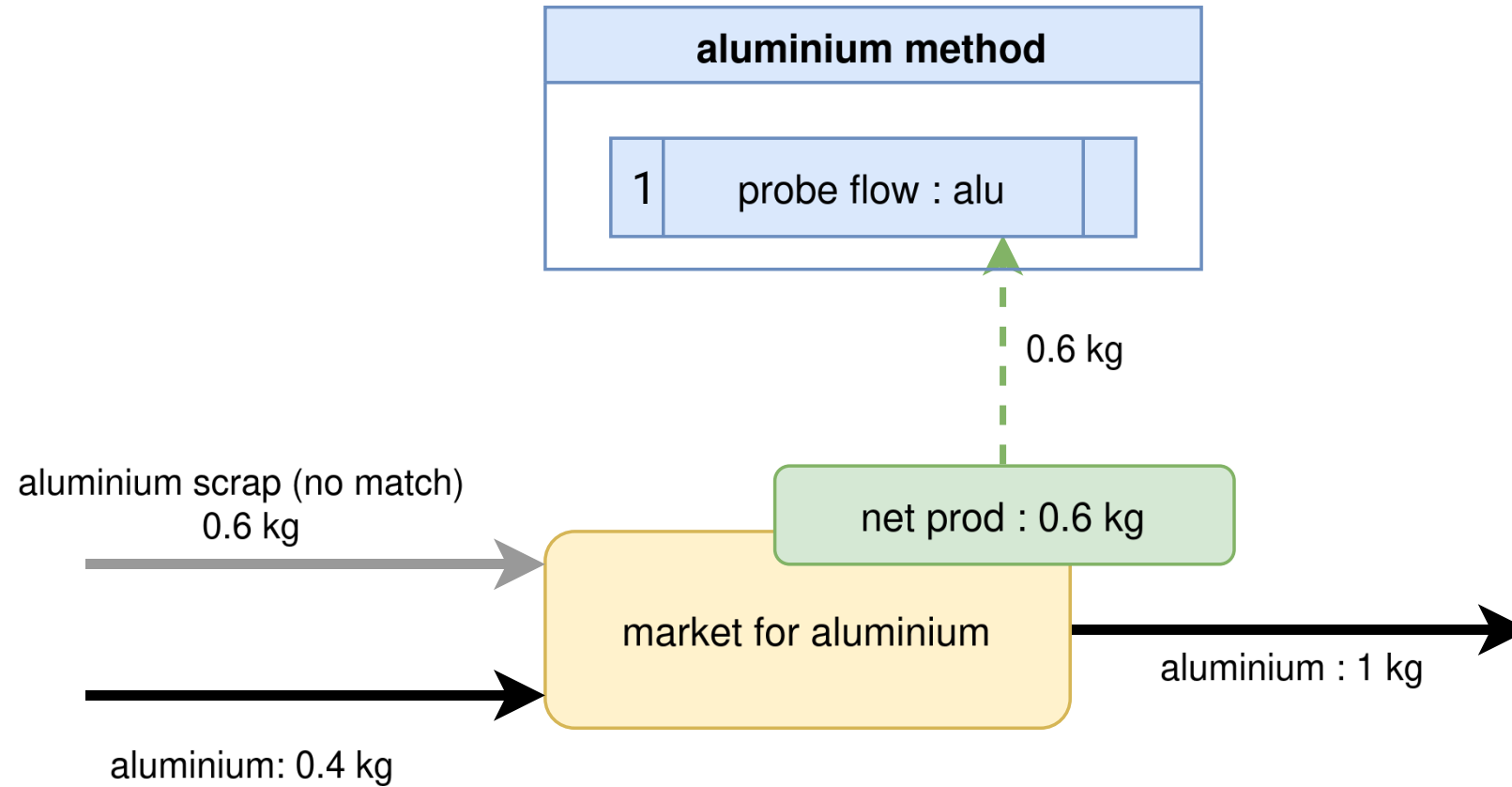
PRINCIPLE



PRINCIPLE



PRINCIPLE



PRINCIPLE

- Matching on **reference product & unit** rather than activity name
 - **Net production** prevents double counting
 - Basic regionalization
 - The setup checks "impacts" of matching activities ≈ 1.0
-
- Principle similar to **T-reX**.
Matching on **ref products + net production** = more generic



SETUP

Requirements

Python 3.12, Brightway 2.5

A project with ecoinvent (tested on 3.11)

Published [on pypi](#) :

> pip install lca_supplymat

```
from lca_supplymat import setup_lca_supply_mat

setup_lca_supply_mat(
    target_db="ecoinvent-3.11-cutoff",
    yaml_file="path.yaml",      # optional: load 23 definition from the package by default
)
```

Full notebook [here](#)

HOW TO DEFINE A METHOD ?

By default lca_supplymat installs [23 pre defined methods](#).

You may also provide your own **YAML file** with **custom definitions** with patterns :

```
concrete:
  matchings: ["*concrete*"]
  excludes: ["*wastewater*", "waste*"]
  unit: cubic meter
```

DEMO



CONCLUSION

RESULTS

- A new (pseudo) LCIA method integrated in brightway framework > fast set-up & user-friendly
- A set of 23 available methods + easy implementation of user-defined methods

Potential uses of this new metric on background/foreground inventories

- Extract material quantity from LCI > **WARNING !!** The quality of the results is representative of the foreground and background inventory datasets > **use results with caution !**
- New lens to explore LCI data to compare them not only on LCI data / traditional impacts
- Helper to build foreground LCI data to quickly assess material quantities
- Compare material consumption from foreground vs background
- ...

THANK YOU FOR YOUR ATTENTION

THANK YOU TO BRIGHTCON AMAZING ORGANIZERS !

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BACK UP SLIDES DEMO

Activity Browser - lca_supplymat_test

Project Database Impact categories Calculate View Help

Databases

name	records	depends
ecoinvent-3.11-biosphere	9795	
ecoinvent-3.11-cutoff	25412	ecoinvent-3.11-biosphere
lca_supplymat	110	ecoinvent-3.11-biosphere lca_supplymat

Databases Impact Categories Calculation Setups

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The Activity Browser



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Activity Browser - lca_supplymat_test

Project Database Impact categories Calculate View Help

Databases

name	records	depends
ecoinvent-3.11-biosphere	9795	
ecoinvent-3.11-cutoff	25412	ecoinvent-3.11-biosphere, lca_supplymat
lca_supplymat	110	

Databases Impact Categories Calculation Setups

lca_supplymat

concrete

name	categories	unit
concrete [CH]	lca_supplymat > concrete	cubic meter
concrete [BR]	lca_supplymat > concrete	cubic meter
concrete [CA-QC]	lca_supplymat > concrete	cubic meter
concrete [GLO]	lca_supplymat > concrete	cubic meter
concrete [PE]	lca_supplymat > concrete	cubic meter
concrete [CO]	lca_supplymat > concrete	cubic meter
concrete [RoW]	lca_supplymat > concrete	cubic meter
concrete [AT]	lca_supplymat > concrete	cubic meter
concrete [ZA]	lca_supplymat > concrete	cubic meter
concrete [IN]	lca_supplymat > concrete	cubic meter
concrete [North America witho...	lca_supplymat > concrete	cubic meter

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Activity Browser - lca_supplymat_test

Project Database Impact categories Calculate View Help

Impact Categories

Quick Search

	name	unit	num_cfs
▶	ecoinvent-3.11		
▼	lca_supplymat		
▼	total		
▼	concrete		
	concrete	cubic meter	11.0
▶	cement		
▶	gravel		
▶	steel-iron		
▶	aluminium		
▶	copper		
▶	rare-earths-oxide		
▶	titanium		
▶	zirconium-oxide		
▶	platinum		
▶	iridium		
▶	nickel-class1		
▶	silicon-metal		
▶	silver		
▶	lithium-carbonate		
▶	lithium-hydroxide		
▶	cobalt-sulfate		
▶	nickel-sulfate		
▶	manganese-sulfate		
▶	graphite		
▶	graphite-nat		
▶	zirconium-sponge		
▶	iron-scrap		

Databases Impact Categories Calculation Setups

Welcome - Parameters - Settings -

Welcome to

The Activity Browser

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[EEROS DATA INIT 2](#)

[default](#)

[name-project](#)

[RTE eolien](#)

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Impact Categories

Quick Search

name	unit	num_cfs
------	------	---------

ecoinvent-3.11

lca_supplymat

total

concrete

concrete	cubic meter	11.0
----------	-------------	------

- ▶ cement
- ▶ gravel
- ▶ steel-iron
- ▶ aluminium
- ▶ copper
- ▶ rare-earths-oxide
- ▶ titanium
- ▶ zirconium-oxide
- ▶ platinum
- ▶ iridium
- ▶ nickel-class1
- ▶ silicon-metal
- ▶ silver
- ▶ lithium-carbonate
- ▶ lithium-hydroxide
- ▶ cobalt-sulfate
- ▶ nickel-sulfate
- ▶ manganese-sulfate
- ▶ graphite
- ▶ graphite-nat
- ▶ zirconium-sponge
- ▶ iron-scrap

Welcome - Parameters - Settings - Characterization Factors x

lca_supplymat | total | concrete x

Name: lca_supplymat | total | concrete

Edit Impact Category

Unit: cubic meter

name	categories	database	amount	unit	uncertain
🌐 concrete [AT]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [BR]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [CA-QC]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [CH]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [CO]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [GLO]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [IN]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [North America without Quebec]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [PE]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [RoW]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	
🌐 concrete [ZA]	lca_supplymat > concrete	lca_supplymat	1.0	cubic meter	

```
1 concrete:
2   matchings: ["*concrete*"]
3   excludes: ["*wastewater*", "waste*"]
4   unit: cubic meter
```

Databases

name	records	depends	def
ecoinvent-3.11-biosphere	9795		uns
ecoinvent-3.11-cutoff	25412	ecoinvent-3.11-biosphere, lca_supplymat	uns
lca_supplymat	110		uns

Databases Impact Categories Calculation Setups

ecoinvent-3.11-cutoff

Quick Search

name	product	unit
concrete production, 25MPa, for building construction, with...	concrete, 2...	cubic

Welcome Parameters Settings LCA Setup Activity Details LCA Results

concrete production, 25MPa, for building construction, with cement, CP III market for concrete, 25MPa

concrete production, 25MPa, for building construction, with cement, CP III

BR ecoinvent-3.11-cutoff

Flows Description Graph Parameters Consumers Data

Output:

amount	unit	product	producer	location	database	categories
1.0	cubic meter		concrete [BR]	GLO	lca_supplymat	lca_supplymat, concrete
89.946	kilogram	waste concrete	market for waste concrete	RoW	ecoinvent-3.11-cutoff	
0.161	cubic meter	wastewater from concrete prod...	market for wastewater from concrete production	RoW	ecoinvent-3.11-cutoff	
0.0499	kilogram	waste mineral oil	market for waste mineral oil	RoW	ecoinvent-3.11-cutoff	
0.0345	kilogram	waste rubber, unspecified	market for waste rubber, unspecified	RoW	ecoinvent-3.11-cutoff	
0.0238	kilogram	waste steel	market for waste steel	RoW	ecoinvent-3.11-cutoff	
1.0	cubic meter	concrete, 25MPa	concrete production, 25MPa, for building constru...	BR	ecoinvent-3.11-cutoff	

Input:

amount	unit	product	producer	location	categories	database	formula
4.57e-07	unit	concrete mixing factory	market for concrete mixing factory	GLO		ecoinvent-3.11-...	
0.0238	kilogram	steel, low-alloyed, hot rolled	market for steel, low-alloyed, hot rolled	GLO		ecoinvent-3.11-...	
0.0345	kilogram	synthetic rubber	market for synthetic rubber	GLO		ecoinvent-3.11-...	
0.0499	kilogram	lubricating oil	market for lubricating oil	RoW		ecoinvent-3.11-...	
0.161	cubic meter		Water, unspecified natural origin		natural resourc...	ecoinvent-3.11-...	
1.85802	kilogram	chemical, organic	market for chemical, organic	GLO		ecoinvent-3.11-...	$1,79 \cdot (1 + \text{loss})$
2.99	kilowatt hour	electricity, medium voltage	market group for electricity, medium voltage	BR		ecoinvent-3.11-...	
5.01	megajoule	diesel, burned in building machine	market for diesel, burned in building machine	GLO		ecoinvent-3.11-...	
190.992	kilogram	tap water	market for tap water	BR		ecoinvent-3.11-...	$184 \cdot (1 + \text{loss})$
316.59	kilogram	cement, CP III	market for cement, CP III	BR		ecoinvent-3.11-...	$\text{cement} \cdot (1 + \text{loss})$
746.322	kilogram	sand	market for sand	BR		ecoinvent-3.11-...	$\text{sand} \cdot (1 + \text{loss})$
1230.03	kilogram	gravel, crushed	market for gravel, crushed	BR		ecoinvent-3.11-...	$\text{gravel} \cdot (1 + \text{loss})$

Databases

name	records	depends	def
ecoinvent-3.11-biosphere	9795		uns
ecoinvent-3.11-cutoff	25412	ecoinvent-3.11-biosphere, lca_supplymat	uns
lca_supplymat	110		uns

Databases Impact Categories Calculation Setups

ecoinvent-3.11-cutoff

Quick Search

name	product	unit
concrete production, 25MPa, for building construction, with...	concrete, 2...	cubic

market for concrete, 25MPa

BR ecoinvent-3.11-cutoff

Flows Description Graph Parameters Consumers Data

Output:

amount	unit	product	producer	location	database	categories	formula	comment	uncert
1.0	cubic meter	concrete, 25MPa	market for concrete, 25MPa	BR	ecoinvent-3.11-cutoff				

No concrete biosphere flow thanks to the algorithm

Input:

amount	unit	product	producer	location	database	category
0.03208607023...	cubic meter	concrete, 25MPa	concrete production, 25MPa, for building construction, with cement, CP-II-F	BR	ecoinvent-3.11-cutoff	
0.06622287093...	cubic meter	concrete, 25MPa	concrete production, 25MPa, for building construction, with cement, CP-II-E	BR	ecoinvent-3.11-cutoff	
0.90169105883...	cubic meter	concrete, 25MPa	concrete production, 25MPa, for building construction, with cement, CP III	BR	ecoinvent-3.11-cutoff	
57.9465	ton kilometer	transport, freight, lorry, >32 metric ...	market for transport, freight, lorry, >32 metric ton, diesel, EURO 3	BR	ecoinvent-3.11-cutoff	

Impact Categories

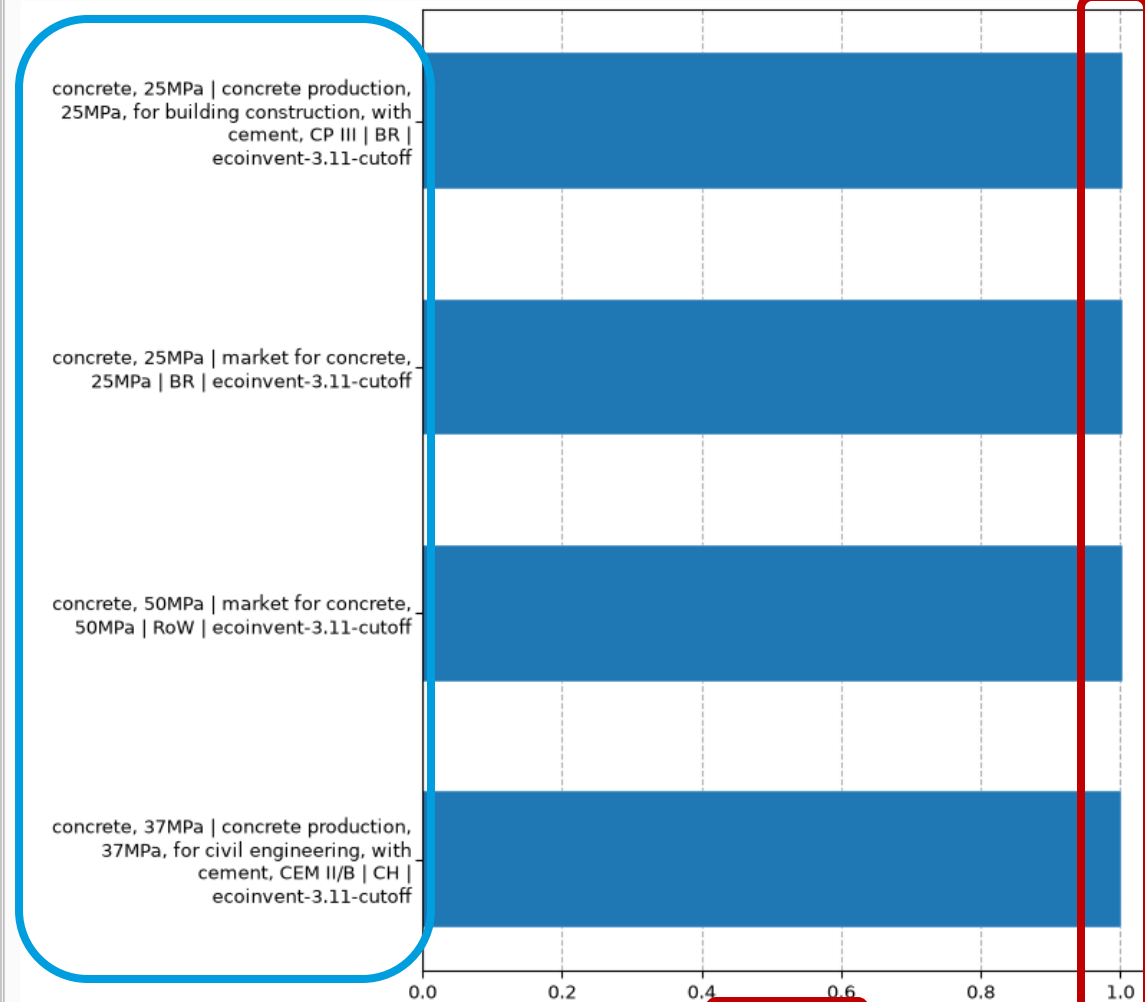
Quick Search

name	unit	num_cfs
ecoinvent-3.11		
lca_supplymat		
total		
concrete		
concrete	cubic meter	11.0
cement		
gravel		
steel-iron		
steel-iron	kilogram	8.0
aluminium		
copper		
rare-earths-oxide		
titanium		
zirconium-oxide		
platinum		
iridium		
nickel-class1		
silicon-metal		
silver		
lithium-carbonate		
lithium-hydroxide		
cobalt-sulfate		
nickel-sulfate		
manganese-sulfate		
graphite		
graphite-nat		
zirconium-sponge		
iron-scrap		

LCA scores

Compare: Reference Flows

Impact Category: ('lca_supplymat', 'tc')

☒ Plot ☐ Table ☐ Relative ☒ Absolute ☒ Horizontal ☒ Full labels

Plot:

Copy

.png

.svg

Table:

Copy

.csv

.xlsx

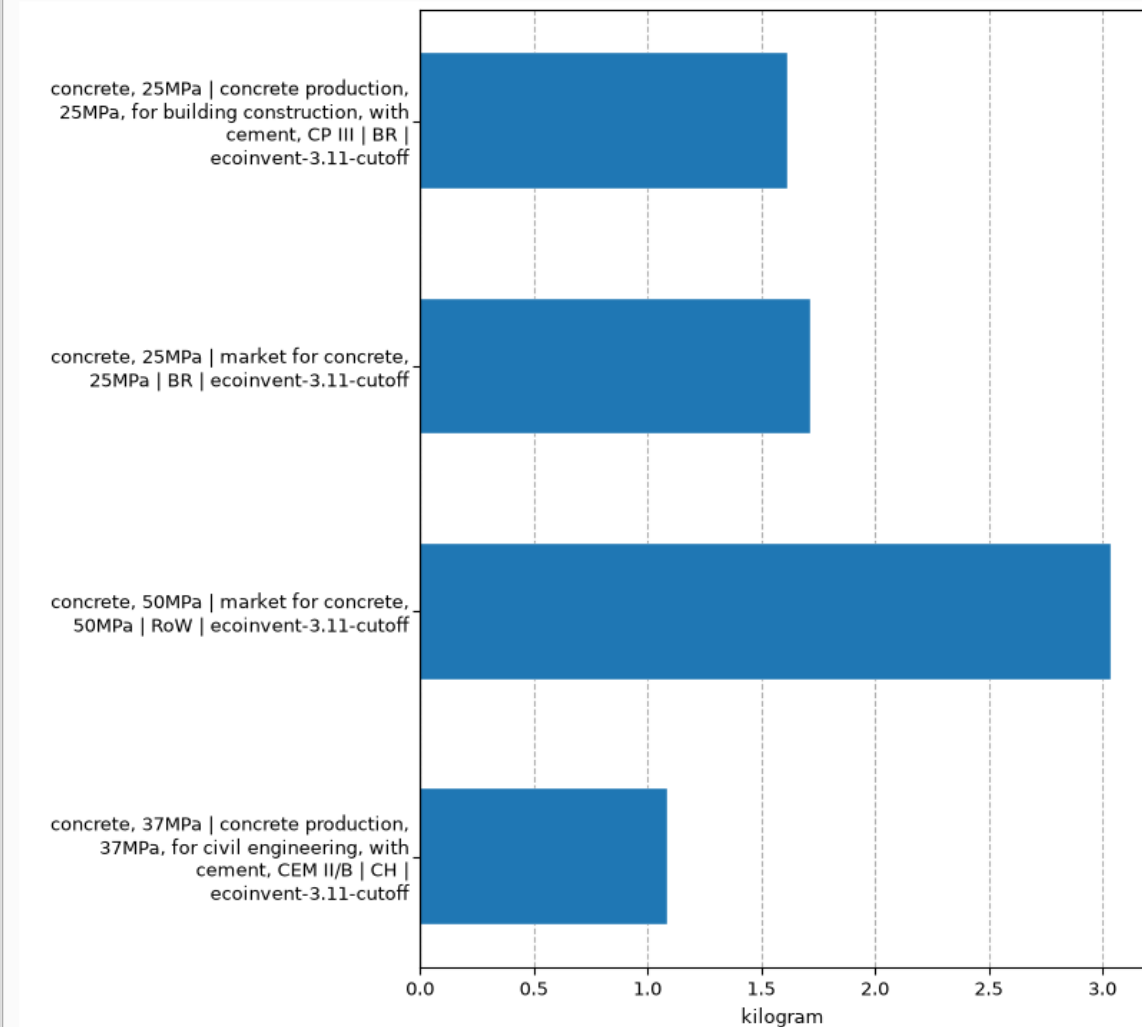
Impact Categories

Quick Search

name	unit	num_cfs
ecoinvent-3.11		
lca_supplymat		
total		
concrete		
concrete	cubic meter	11.0
cement		
gravel		
steel-iron		
steel-iron	kilogram	8.0
aluminium		
copper		
rare-earths-oxide		
titanium		
zirconium-oxide		
platinum		
iridium		
nickel-class1		
silicon-metal		
silver		
lithium-carbonate		
lithium-hydroxide		
cobalt-sulfate		
nickel-sulfate		
manganese-sulfate		
graphite		
graphite-nat		
zirconium-sponge		
iron-scrap		

LCA scores

Compare: Reference Flows Impact Category: ('lca_supplymat', 'tc')

☒ Plot ☐ Table ☐ Relative ☒ Absolute ☒ Horizontal ☒ Full labels

Plot: Copy .png .svg Table: Copy .csv .xlsx

STEEL

lca_supplymat, total, steel-iron

Impact Categories

Quick Search

name	unit	num_cfs
ecoinvent-3.11		
lca_supplymat		
total		
concrete		
concrete	cubic meter	11.0
cement		
gravel		
steel-iron		
steel-iron	kilogram	8.0
aluminium		
copper		
rare-earths-oxide		
titanium		
zirconium-oxide		
platinum		
iridium		
nickel-class1		
silicon-metal		
silver		
lithium-carbonate		
lithium-hydroxide		
cobalt-sulfate		
nickel-sulfate		
manganese-sulfate		
graphite		
graphite-nat		
zirconium-sponge		
iron-scrap		

Welcome Parameters Settings LCA Setup x Activity Details x LCA Results x Characterization Factors x

concrete [14:26] x elec [14:12] x steel [14:22] x copper [14:20] x transport [14:42] x concrete [16:15] x

Inventory LCA scores EF Contributions Process Contributions Tree Sankey Monte Carlo Sensitivity Analysis

LCA scores

Compare: Reference Flows

Impact Category: ('lca_supplymat', 'tc')

Plot Table Relative Absolute Horizontal Full labels



It also includes the steel in the upstream supply chain = the steel that is used to produce steel

Calculation Setups

name	functional_units	impact_categories
elec	4	4
transport	6	5
concrete	4	4
copper	3	3
steel	3	3

LCA scores

Compare: Reference Flows

Impact Category: ('lca_supplymat', 'tc')

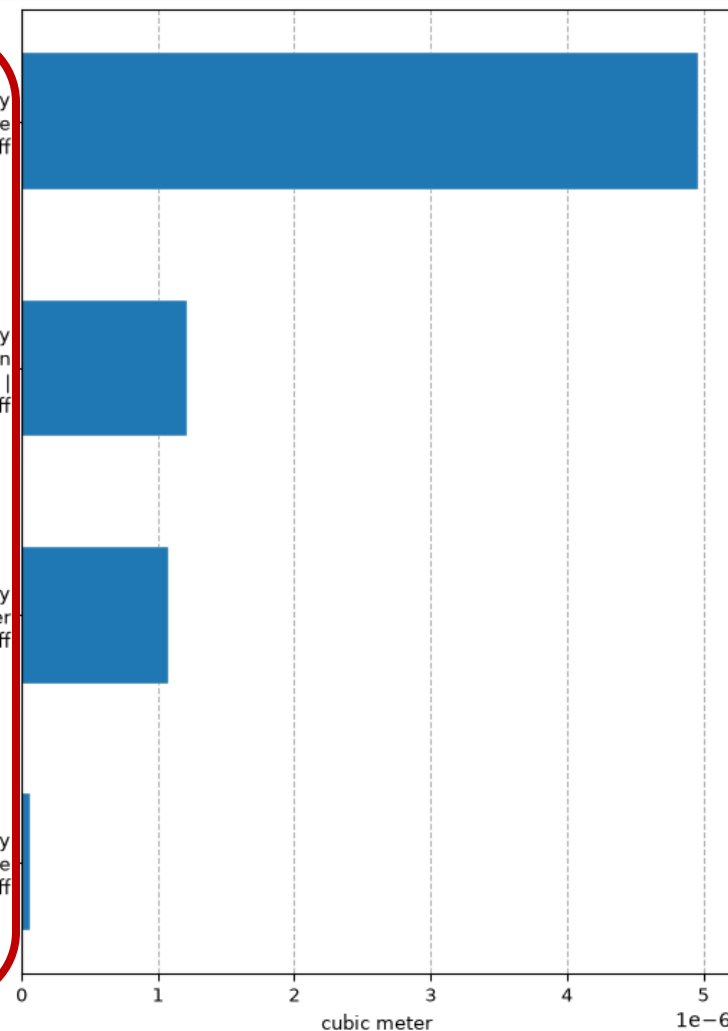
☒ Plot ☐ Table ☐ Relative ☒ Absolute ☒ Horizontal ☐ Full labels

electricity, high voltage | electricity
production, wind, 1-3MW turbine, onshore
| FR | ecoinvent-3.11-cutoff

electricity, low voltage | electricity
production, photovoltaic, 570kWp open
ground installation, multi-Si | FR |
ecoinvent-3.11-cutoff

electricity, high voltage | electricity
production, nuclear, pressure water
reactor | FR | ecoinvent-3.11-cutoff

electricity, high voltage | electricity
production, hydro, reservoir, alpine
region | FR | ecoinvent-3.11-cutoff



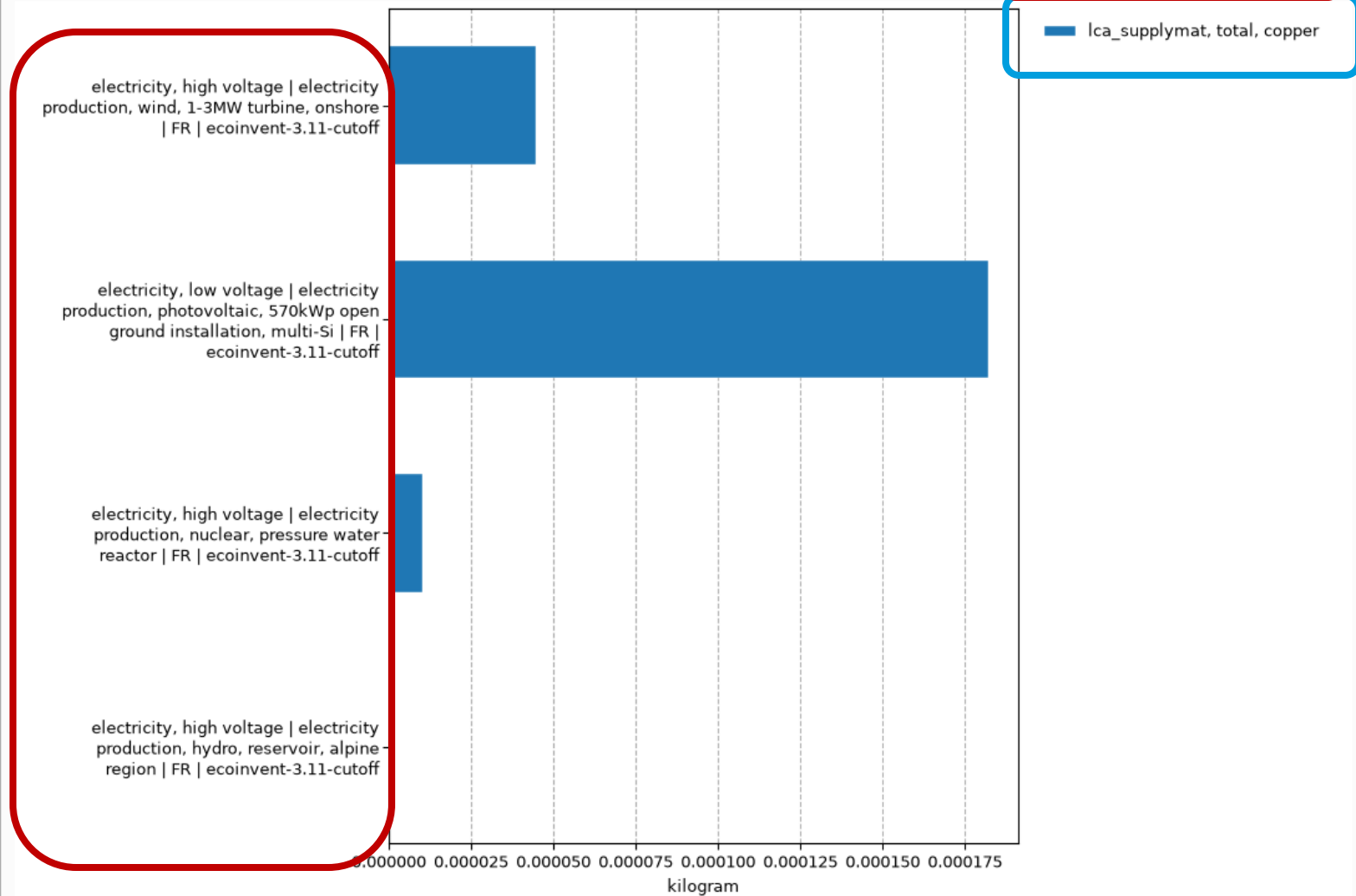
lca_supplymat, total, concrete

Calculation Setups

name	functional_units	impact_categories
elec	4	4
transport	6	5
concrete	4	4
copper	3	3
steel	3	3

LCA scores

Compare: Reference Flows Impact Category: ('lca_supplymat', 'tc')

☒ Plot ☐ Table ☐ Relative ☒ Absolute ☒ Horizontal ☐ Full labels

Calculation Setups

name	functional_units	impact_categories
elec	4	4
transport	6	5
concrete	4	4
copper	3	3
steel	3	3

- Electricity production, PV
- Concrete method

Sankey

Reference flow: electricity, low volta Impact category: ('lca_supplymat', 'tc

Cutoff: 6.000 %

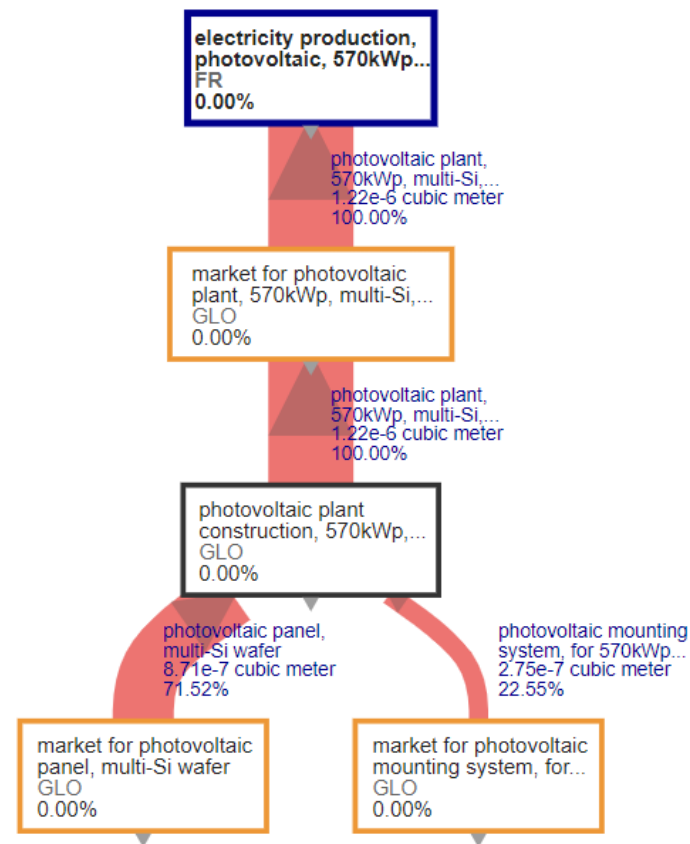
Adjust to: Tier

2

Adjust

Aggregate by: None

Color by: Direct impact

Layout Top to bottom ☐ Minimap ☒ Flows ☒ absolute ☒ relative

Shown: 5 processes, 0.0% of direct impacts, max tier 3, 4 flows | Calculated: 5 processes, 0.0% of direct impacts, max tier 3

Show all

Plot:

Copy

.png

.svg